

Sentiment Analysis: corpus-based approaches (cont.) + Information Extraction (intro)

Carolina Scarton

`c.scarton@sheffield.ac.uk`

Text Processing
Department of Computer Science, University of Sheffield
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The 17th BCSWomen Lovelace Colloquium

University of Liverpool

04th April 2024



Poster contests open to all **women and non-binary UG/PGT taught students** of computing in the UK, with categories for:

first year / foundation year

second year/ industrial year

final year UG

MSc / final year MEng/MComp

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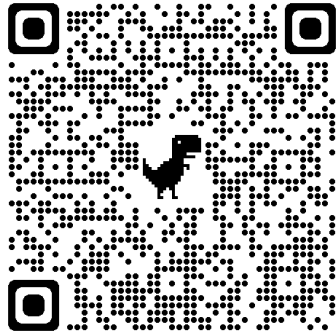


SCAN FOR MORE
INFORMATION

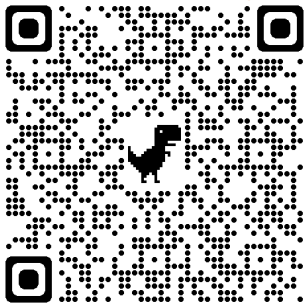
To enter submit a 250 word abstract by 5th February 2024 for a chance to win prizes (£300 for 1st place)

SWiCS
ABSTRACT
WRITING
WORKSHOP

24th November | 1pm - 3pm
Ada Lovelace room, Regents Court



Deadline: 01/12 @ 5pm



<https://forms.gle/Ujm2kTjxLxQGc9f7>

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**Do you have something interesting to say about
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To celebrate the diversity in community, we particularly invite
students that are:

- Female / non-binary / LGBTQ+
- First of their family to attend University
- Part of the UoS Access+ programme

**Submit a short video for a chance to win
one of five £100 Amazon voucher!!!**

Learning objectives

- ▶ Explain the Binary Naïve Bayes model
- ▶ Apply your knowledge about Naive Bayes for Sentiment Analysis
- ▶ Evaluate classifiers for text classification
- ▶ Explain the Information Extraction task
- ▶ Name the main subtasks within Information Extraction
- ▶ Define the main approaches used for Information Extraction

[Sentiment Analysis]

Naive Bayes recap

Corpus-based approaches: Naive Bayes

Bayes classifier:

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} \frac{p(T|s_i)p(s_i)}{p(T)}$$

Corpus-based approaches: Naive Bayes

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Since **evidence** $\mathbf{p}(\mathbf{T})$ is independent of s_i , we can ignore it

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$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(T|s_i)p(s_i)$$

- ▶ $p(s_i|T)$ is the **posterior probability**
- ▶ $p(T|s_i)$ is the **likelihood**
- ▶ $p(s_i)$ is the **prior probability**

Corpus-based approaches: Naive Bayes

Prior probability

Compute prior probability by simple relative frequency

$$p(s_i) = \frac{\text{count}(s_i)}{\sum_{j=0}^J \text{count}(s_j)}$$

with J the number of different classes and $\text{count}(\cdot)$ is the counting function

Corpus-based approaches: Naive Bayes

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Likelihood

Naive assumption: **features** are **independent**

$$p(T|s_i) = p(t_1, t_2, \dots, t_N|s_i) \approx \prod_{j=1}^N p(t_j|s_i)$$

$\Rightarrow$ product of probabilities of each **feature** value of text occurring with class s_i

Doc	Words	Class
1	A <u>sensitive</u> , <u>moving</u> , <u>brilliant</u> work	Positive
2	An <u>edgy</u> thriller that delivers a <u>surprising</u> punch	Positive
3	A <u>sensitive</u> , <u>insightful</u> , <u>flamboyant</u> film	Positive
4	Neither <u>revelatory</u> nor truly <u>edgy</u> – merely crassly <u>flamboyant</u> and comedically <u>labored</u>	Negative
5	<u>Unlikable</u> , <u>uninteresting</u> , <u>unfunny</u> , and completely, utterly <u>inept</u>	Negative
6	A sometimes <u>incisive</u> and <u>sensitive</u> portrait, targeting <u>sensitive</u> topics, that is undercut by its <u>awkward</u> structure	Negative
7	It's a sometimes <u>interesting</u> remake that doesn't compare to the <u>brilliant</u> original	Negative

Classify the following sentence using a Naive Bayes classifier based on the above training data (without smoothing).

Doc	Words	Class
9	<u>Amazing</u> movie	???

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

2. Consider other **feature representations**?

- ▶ Instead of frequencies → binary count
- ▶ 1 if feature appears in a segment, 0 otherwise

		NB Counts		Binary Counts		
		+	-	+	-	
Four original documents:						
-	it was pathetic the worst part was the boxing scenes	and	2	0	1	0
		boxing	0	1	0	1
		film	1	0	1	0
-	no plot twists or great scenes	great	3	1	2	1
+	and satire and great plot twists	it	0	1	0	1
+	great scenes great film	no	0	1	0	1
		or	0	1	0	1
		part	0	1	0	1
		pathetic	0	1	0	1
		plot	1	1	1	1
		satire	1	0	1	0
		scenes	1	2	1	2
+	and satire great plot twists	the	0	2	0	1
+	great scenes film	twists	1	1	1	1
		was	0	2	0	1
		worst	0	1	0	1

Figure 4.3 An example of binarization for the binary naive Bayes algorithm.

[Jurafsky and Martin, 2021]

Naive Bayes for Sentiment Analysis - Binary example

Consider the following **training set**:

doc	“good”	“poor”	“great”	(class)
d1.	3	0	3	pos
d2.	0	1	2	pos
d3.	1	3	0	neg
d4.	1	5	2	neg
d5.	0	2	0	neg

[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

”A **good**, **good** plot and **great** characters, but **poor** acting.”

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[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

”A **good**, **good** plot and **great** characters, but **poor** acting.”

Standard NB:

$$\begin{array}{l|l} p(\text{good}|\text{positive}) = 3/9 = 1/3 & p(\text{good}|\text{negative}) = 1/14 \\ p(\text{great}|\text{positive}) = 5/9 & p(\text{great}|\text{negative}) = 2/14 = 1/7 \end{array}$$

Naive Bayes for Sentiment Analysis - Binary example

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[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

“A **good**, **good** plot and **great** characters, but **poor** acting.”

Binary NB:

$$\begin{array}{l|l} p(\text{good}|\text{positive}) = 1/4 & p(\text{good}|\text{negative}) = 2/6 = 1/3 \\ p(\text{great}|\text{positive}) = 2/4 = 1/2 & p(\text{great}|\text{negative}) = 1/6 \end{array}$$

Exercise

Doc	Words	Class
1	A <u>sensitive</u> , <u>moving</u> , <u>brilliant</u> work	Positive
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Classify the following sentence using a **Binary Naive Bayes** classifier based on the above training data **without smoothing**.

Doc	Words	Class
8	A <u>flamboyant</u> romcom, <u>sensitive</u> but <u>awkward</u> at times.	???

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

3. Consider other **algorithms**?

- ▶ Maximum Entropy (MaxEnt), Support Vector Machines (SVM), neural networks
- no assumption of statistical independence among features
- more complex but tend to do better

→ Take a look at Chapter 5 of [Jurafsky and Martin, 2021] for **Text Classification with Logistic Regression**

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- ▶ Different **evaluation metrics**
 - to be used in different scenarios
- ▶ Evaluation in the **development set**
 - model selection
- ▶ Evaluation in the **test set**
 - generalisation

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

		Predicted class	
		Positive	Negative
Actual class	Positive	<i>TP</i>	<i>FN</i>
	Negative	<i>FP</i>	<i>TN</i>

TP = true positives

TN = true negatives

FP = false positives

FN = false negatives

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- Different **evaluation metrics** → commonly used for classification

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2 * TP + FP + FN}$$

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- ▶ Different **evaluation metrics** → commonly used for classification
 - ▶ Accuracy → not suitable for **imbalanced data problems** (majority class bias)

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- ▶ Different **evaluation metrics** → commonly used for classification
 - ▶ Accuracy → not suitable for **imbalanced data problems** (majority class bias)
 - **F1** more reliable → for **binary classification only!**

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- ▶ Different **evaluation metrics** → non-binary approaches
 - ▶ $F1$ is still reliable if data is balanced

For imbalanced problems:

$$\text{Sensitivity} = \text{Recall} = \text{TPR} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{FP + TN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

$$F\beta = \frac{(1 + \beta^2) * \text{Precision} * \text{Recall}}{(\beta^2 * \text{Precision}) + \text{Recall}}$$

If $\beta > 1 \rightarrow$ weighs recall higher

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- ▶ **No test set???**

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

- ▶ Different **evaluation metrics**
- a reliable evaluation is done with multiple metrics!
- ▶ **No test set???**
- **cross-validation:** divide the training set in k folds
 - ▶ for each unique group:
 - ▶ use the group as test data
 - ▶ use the $k - 1$ groups as training data
 - ▶ training the model in $k - 1$ and evaluate in the test set
 - ▶ save the evaluation score (Accuracy, F1, etc).
 - ▶ **Final score:** average value considering all k evaluations

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2 * TP + FP + FN}$$

Calculate **macro-F1** and **accuracy**:

Classifier 1:

	Predicted class	
	Positive	Negative
Actual class	Positive	TP=90 FN=10
	Negative	FP=40 TN=5

Naive Bayes for Sentiment Analysis → Evaluation

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Classifier 1:

		Predicted class	
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Classifier 2:

		Predicted class	
		Positive	Negative
Actual class	Positive	TP=70	FN=30
	Negative	FP=5	TN=40

Naive Bayes for Sentiment Analysis → Evaluation

How can we evaluate the performance of a classifier?

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Which classifier is the best?

Naive Bayes classifier → other applications

Name other applications for which similar approaches can be applied

[Information Extraction]

Introduction

- ▶ **Introduction to Information Extraction**
 - ▶ **Definition + contrast with IR**
 - ▶ **Example Applications**
 - ▶ **Overview of Tasks**
 - ▶ **Overview of Approaches**
 - ▶ Evaluation + Shared Task Challenges
 - ▶ Brief(est) history of IE
- ▶ **Named Entity Recognition**
 - ▶ Task
 - ▶ Approaches: Rule-based, Supervised Learning
 - ▶ Entity Linking
- ▶ **Relation Extraction**
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Information Extraction

Definition

Information Extraction is the task of **identifying information** about predefined classes of **entitites**, **relationships** or **events** and record it in a **structured form**

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Information Extraction is the task of **identifying information** about predefined classes of **entitites**, **relationships** or **events** and record it in a **structured form**

Other definitions:

- ▶ The activity of populating a structured information repository from an unstructured information source
- ▶ The activity of creating a semantically annotated text collection (→ **semantic web**)

Information Extraction

Why doing **information extraction**?

- ▶ searching or analysis using conventional database queries
 - Difficult to search directly in the text because it lacks structure.

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Information Extraction

Why doing **information extraction**?

- ▶ searching or analysis using conventional database queries
 - Difficult to search directly in the text because it lacks structure.
- ▶ data mining
- ▶ summarisation (eventually in another language)
- ▶ construct indexes into/within/between large quantity of text

Information Extraction

- ▶ **Entities:** persons, organisations, locations, times, etc. →
Named Entities
- ▶ **Relationships:** links between entities, etc.
- ▶ **Events:** succession events, etc.

Information Extraction

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Named Entities
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The **structured form** can be implemented either as a database or form (slot filling) or by using XML tags (tagging)

Information Extraction: Example

Who's News: @ Burns Fry Ltd.

04/13/94 WALL STREET JOURNAL (J), PAGE B10 <CO>
BURNS FRY Ltd. (Toronto) – Donald Wright , 46
years old, was named executive vice president and
director of fixed income at this brokerage firm. Mr. Wright
resigned as president of Merrill Lynch Canada Inc. , a unit
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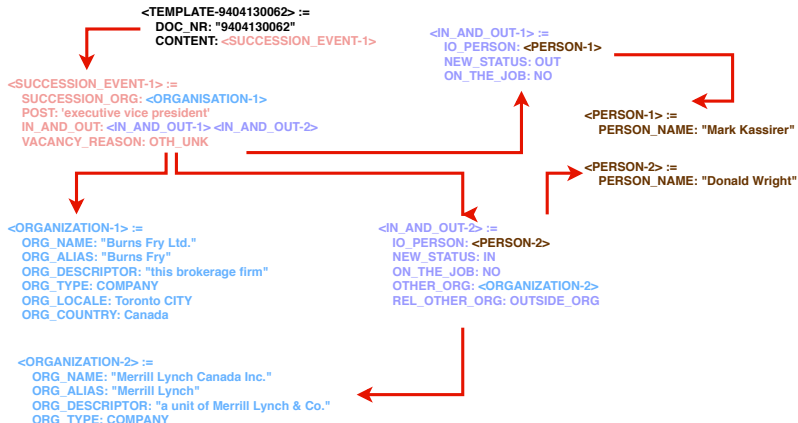
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Information Extraction: Filled template



Information Extraction vs. Information Retrieval

IR Task:

- ▶ Given a **document collection** and a **user query**
- ▶ Returns a (ranked) **list of documents relevant to the user query**

Information Extraction vs. Information Retrieval

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Strengths:

- ▶ Can search huge document collections very rapidly
- ▶ Insensitive to genre and domain of the texts
- ▶ Relatively straightforward to implement
 - ▶ challenges scaling to huge, dynamic document collections, e.g. the web

Information Extraction vs. Information Retrieval

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Weaknesses

- ▶ Documents are returned rather than information/answers
 - ▶ user must further read texts to extract information
 - ▶ output is unstructured so limited possibilities for further processing
- ▶ Frequently not discriminative enough (“14,100,000 documents match your request”)

Information Extraction vs. Information Retrieval

IE Task:

- ▶ Given a **document collection** and a **predefined set of entities, relations and/or events**
- ▶ Returns a **structured representation** of all mentions of the specified entities, relations and/or events

Information Extraction vs. Information Retrieval

IE Task:

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- ▶ Can feed other powerful applications (databases, semantic indexing engines, data mining tools)

Information Extraction vs. Information Retrieval

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- ▶ Returns a **structured representation** of all mentions of the specified entities, relations and/or events

Strengths:

- ▶ Extracts **facts** from texts, not just texts from text collections
- ▶ Can feed other powerful applications (databases, semantic indexing engines, data mining tools)

Weaknesses:

- ▶ Systems tend to be genre/domain specific and porting to new genres and domains can be time-consuming/requires expertise
- ▶ Limited accuracy
- ▶ Computationally demanding, so performance issues on very large collections

Information Extraction: example applications

Information Extraction: example applications

- ▶ Google News uses Named Entity Recognition for its “In the News” feature
- ▶ Scrapping web pages to build structured databases of job postings, apartment rentals, seminar announcements, etc.
- ▶ Assisting biomedical database curators by extracting biomedical entities and relations from the scientific literature prior to entry in a human-maintained database (e.g. Flybase)
- ▶ Assisting companies in competitor intelligence gathering, e.g. management or researcher succession events, new product or project announcements, etc.

Information Extraction: Overview

- ▶ **Introduction to Information Extraction**

- ▶ Definition + contrast with IR
- ▶ Example Applications
- ▶ **Overview of Tasks**
- ▶ Overview of Approaches
- ▶ Evaluation + Shared Task Challenges
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Information Extraction: Overview of tasks

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The **set of types** is determined in advance (e.g. organisation, person, date, etc...)

Information Extraction: Overview of tasks

Entity Extraction/Named Entity Recognition (NER)

Task: Identify the **extent** and the **type** of each textual mention of an entity

The **set of types** is determined in advance (e.g. organisation, person, date, etc...)

Cable and Wireless	today announced ...	Extent: 0-3 ; Type = ORG
IBM	and Microsoft	today announced ...
		Extent: 0-1 ; Type = ORG
		Extent: 2-3 ; Type = ORG
John Lewis	hired ...	Extent: 0-2 ; Type = ORG
Theresa May	hired.	Extent: 0-2 ; Type = PER

Information Extraction: Entity extraction

Types of entities addressed by IE systems include:

- ▶ **Named individuals**

- ▶ Organisations (ORG), persons (PER), books, films, ships, restaurants ...

- Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**

- Barack Obama was the 44th president ...; Extent: **0-2** ; Type = **PER**

Information Extraction: Entity extraction

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► Named individuals

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- Cable and Wireless today announced ...; Extent: **0-3** ; Type = **ORG**
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- Geo-Political entities (GPE), locations (LOC)
- The Mont Blanc intersects France, Italy and Switzerland. ; Extent: **1-3** ; Type = **LOC**
- The Mont Blanc intersects France , Italy and Switzerland . ; Extent: **4-5** ; Type = **GPE**

Information Extraction: Entity extraction

Types of entities addressed by IE systems include:

► Named individuals

- Organisations (ORG), persons (PER), books, films, ships, restaurants ...
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► Times: temporal expressions dates, times of day

- Let's meet at 2pm next Friday ...; Extent: **3-4** ; Type = **TIME**
- Let's meet at 2pm next Friday ...; Extent: **5-6** ; Type = **DATE**

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► Measures: monetary expressions, distances/sizes, weights ...

- This watch costs £35 ...; Extent: **3-4** ; Type = **MONEY**

Information Extraction: Entity extraction: coreference

Coreference

Different textual expressions that refer to the same real world entity are said to **corefer**. **Coreference Task**: link together all textual references to the same **real world entity**,

Information Extraction: Entity extraction: coreference

Coreference

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Multiple references to the same entity in a text are **rarely made using the same string**:

- ▶ Pronouns: **Tony Blair == he**
- ▶ Names/definite descriptions: **Tony Blair == the Prime Minister**
- ▶ Abbreviated forms: **Theresa May == May; European Union == EU**
- ▶ Orthographic variants: **alpha helix == alpha-helix == α -helix == a-helix**

Can be seen as a separate task or as part of entity extraction task

Information Extraction: Relation extraction

Relation Extraction

Identify all assertions of relations between entities

Relation Extraction

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May be divided into two subtasks:

- ▶ **Relation detection:** find pairs of entities between which a relation holds
- ▶ **Relation classification:** determine the type of a previously detected relation

Information Extraction: Relation extraction

Relation Extraction

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Example: LOCATION_OF holding between

- ▶ ORGANISATION and GEOPOLITICAL_LOCATION
- ▶ medical INVESTIGATION and BODY_PART
- ▶ GENE and CHROMOSOME_LOCATION
- ▶ EMPLOYEE_OF holding between PERSON and ORGANISATION
- ▶ PRODUCT_OF holding between ARTIFACT and ORGANISATION

Information Extraction: Relation extraction

Challenges for Relation Extraction

- ▶ Many different ways to **express the same relation**
 - ▶ **Canonical:** [Microsoft]_{ORG} is located in [Redmond]_{LOC}
 - ▶ **Synonyms:** [Microsoft]_{ORG}
is located/based/headquartered in [Redmond]_{LOC}
 - ▶ **Syntactic variations:**
 - ▶ [Microsoft]_{ORG}, the software giant and ... , is based in [Redmond]_{LOC}
 - ▶ [Redmond]_{LOC} -based [Microsoft]_{ORG} ...
 - ▶ [Redmond]_{LOC} 's [Microsoft]_{ORG} ...
 - ▶ [Redmond]_{LOC} software giant [Microsoft]_{ORG} ...

Challenges for Relation Extraction

- Relations often involve **coreference links**

[Bill Gates]_{PER}

is co-founder, technology advisor and board member of
[Microsoft]_{ORG}. **[He]**_{PER} served as chairman of the board until
Feb. 4, 2014. On June 27, 2008, **[Gates]**_{PER} transitioned out of a
day-to-day role in the company to spend more time on his global
health and education work at the [Bill & Melinda Gates
Foundation]_{ORG}.

Information Extraction: Event extraction

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Identify all reports of event instances, typically of a small set of classes

Information Extraction: Event extraction

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Identify all reports of event instances, typically of a small set of classes

May be divided into two subtasks:

- ▶ **Event detection:** find all mentions of events in a text
- ▶ **Event classification:** assign a class to the detected events

Examples

- ▶ National/european elections
- ▶ Management succession events
- ▶ Joint venture/product announcements
- ▶ Terrorist attacks

Information Extraction: Event extraction

Challenges for Event Extraction

Events may be simply viewed as relations. However they are typically complex relations

- ▶ often **temporally situated**, often short duration
- Bolt etched his name in history with a 9.69-second finish **on Aug. 16, 2008.**

Information Extraction: Event extraction

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Banks, businesses, civil society and governments at all levels are to announce initiatives to finance and build a new generation of sustainable cities at the **UN Climate Action Summit** in New York today.

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- ▶ often expressed **across multiple sentences**

Information Extraction: Overview

- ▶ **Introduction to Information Extraction**

- ▶ Definition + contrast with IR
- ▶ Example Applications
- ▶ Overview of Tasks
- ▶ **Overview of Approaches**
- ▶ Evaluation + Shared Task Challenges
- ▶ Briefest history of IE

- ▶ **Named Entity Recognition**

- ▶ Task
- ▶ Approaches: Rule-based, Supervised Learning
- ▶ Entity Linking

- ▶ **Relation Extraction**

- ▶ Task
- ▶ Approaches: Rule-based, Supervised Learning, Bootstrapping, Distant Supervision

Information Extraction: Overview of Approaches

Approaches to IE may be placed into four categories:

1. Knowledge Engineering Approaches
2. Supervised Learning Approaches
3. Bootstrapping Approaches
4. Distant Supervision Approaches

Information Extraction: Knowledge Engineering Approaches

[Mr. Wright]_{PER} , [executive vice president]_{POSITION} of [Merrill Lynch Canada Inc.]_{ORG}

Use **manually authored** rules that can be divided into

- ▶ “deep” – **linguistically inspired** language understanding systems
- ▶ “shallow” – systems engineered to the IE task, typically using **pattern-action rules**

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Pattern: “Mr. \$Uppercase-initial-word”

Action: **add-entity(person(Mr. \$Uppercase-initial-word))**

Pattern: “\$Person, \$Position of \$Organization”

Action: **add-relation(is-employed-by(\$Person,\$Organization))**

Information Extraction: Supervised Learning Approaches

⇒ **Machine Learning systems** trained on manually annotated texts (entities and relations)

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For each entity/relation create a training instance

- ▶ k words either side of an entity mention
 - ▶ k words to the left of entity 1 and to the right of entity 2 plus the words in between
- extract **features**: words, POS, morphology

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- ▶ classifiers that classify tokens as beginning/inside/outside a tag type

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Machine Learning techniques: covering algorithms, HMMs, SVMs

Information Extraction: Bootstrapping Approaches

⇒ A technique for relation extraction that requires only **minimal supervision**

Systems are given

- ▶ seed tuples: e.g. **⟨Microsoft , Redmond⟩**
- ▶ seed patterns: e.g. **[X]_{ORG} is located in [Y]_{LOC}**

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- process iterates until convergence
- (detailed in a later lecture)

Information Extraction: Distant Supervision Approaches

⇒ also called “**weakly labelled**” or “**lightly supervised**” approaches

1. Assumes a (semi-)structured data source, such as
 - ▶ Wikipedia infoboxes (e.g. `PERSON BORN_IN LOCATION/DATE`)
 - ▶ Yeast Protein Database, (e.g. `PROTEIN IS__LOCATED__IN SUBCELLULAR__ LOCATION`)
 - contains tuples of entities standing in the relation of interest
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3. **Labelled data is used to train a standard supervised NER or RE system**
 - See later lecture

Reading material

- ▶ Daniel Jurafsky & James H. Martin (2021).
Chapter 17: Information Extraction.
Speech and Language Processing, 3rd ed. draft.
<https://web.stanford.edu/~jurafsky/slp3/17.pdf>

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

References I

Jurafsky, D. and Martin, J. H. (2021).

Speech and Language Processing, see chapter 4: *Naive Bayes and Sentiment Classification*.